Hybrid Sampling and Machine Learning for Cardiovascular Disease Prediction: Data Leakage in Resample-Before-Split Evaluation and a Leakage-Free Pipeline

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ABSTRACT

Cardiovascular disease (CVD) remains one of the foremost causes of death globally, and machine learning (ML) models trained on population health surveys are widely proposed for early risk screening. Because CVD cases are a minority in such data, studies commonly rebalance the classes with hybrid sampling (HS), here SMOTEENN followed by Random Oversampling (ROS). This study evaluates HS with three deep learning (DL) models (CNN, NN, MLP) and three classical ML models (Decision Tree, SVM, XGBoost) on 229,781 unique records of the 2015 BRFSS survey, under two evaluation protocols. When the whole dataset is resampled before it is split, as in many published pipelines, the models reach 94.7–99.8% test accuracy (Decision Tree 99.8%, XGBoost 98.2%, MLP 98.5%). We show that these scores are an artefact of data leakage: every positive test record has an identical copy in the training set, and the edited-nearest-neighbour step removes 19.9% of the real non-CVD records, precisely the borderline cases, before the test set is drawn. When the test set (6,000 real records) is held out before any sampling, the same models reach only 57.4–70.8% accuracy and an area under the ROC curve (AUC) of 0.57–0.81. We therefore propose a leakage-free pipeline that uses all 21 survey features, class weighting instead of synthetic resampling, regularised models and a decision threshold tuned on a validation set. It improves every model on unseen patients, to 74.9–76.0% accuracy, 78.8–82.8% recall and an AUC of 0.825–0.840 (Decision Tree accuracy 57.4% → 74.9%, XGBoost recall 30.5% → 82.8%), with five-fold cross-validation confirming these values (AUC 0.82–0.84). Simple models such as logistic regression perform on par with deep networks. The results show that the evaluation protocol, not the choice of model, dominates reported performance on imbalanced clinical data, and they provide realistic expectations and a reproducible, open-source baseline for CVD screening from survey data.

Keywords: Cardiovascular Disease, Disease Prediction, Machine Learning, Neural Networks, Hybrid Sampling, Class Imbalance, Data Leakage

1 INTRODUCTION

According to the World Health Organization (WHO), CVDs are a group of conditions that affect the heart and blood vessels, leading to various health issues and accounting for approximately 31% of all global deaths annually [1]. It has been estimated that by 2030, deaths related to CVD will rise significantly, with approximately 23.3 million people dying from this condition [2]. These types of diseases include, but are not limited to, coronary artery disease, myocardial infarctions, angina, stroke, arrhythmias, atherosclerosis, and hypertension. Myocardial infarctions and angina are primarily caused when the arteries become narrowed or blocked, resulting in reduced blood supply to the heart, thereby escalating the risk of CVDs [3, 4]. It has also been found that high blood pressure, along with arterial stiffening, is a major cause of CVD. Additionally, factors such as alcohol use, stress, physical inactivity, obesity and its complications, and diabetes also contribute to the development of CVD [5]. Given the silent yet deadly progression of heart disease, early and accurate diagnosis becomes increasingly challenging. Therefore, early detection and intervention in CVD are crucial, as timely and accurate treatment can help prevent life-threatening complications [6].

Although age, gender, blood pressure, and smoking status have traditionally been employed as fixed factors for predicting CVD, they do not adequately reflect the disease's dynamic nature and complexity [7, 8]. In contrast, techniques in data mining and machine learning (ML) have demonstrated encouraging results in assessing the severity of heart disease and forecasting outcomes, with algorithms such as K-nearest neighbors (KNN), Decision Tree (DT), and Naive Bayes (NB) showing effectiveness [9]. Consequently, advanced analytical approaches are crucial not only for early detection but also for the effective management of the numerous factors linked to CVD. However, deep learning (DL) approaches often encounter challenges concerning data interpretation, transparency, and reliability in medical contexts [10]. To overcome these limitations, hybrid models combining DL with other computational techniques have been developed, offering enhanced explainability, predictive accuracy, and clinical relevance.

A second, less discussed challenge is class imbalance. In population health surveys only a small fraction of respondents report CVD (about 10% in the data used here), so classifiers trained naively tend to predict the majority class. Resampling methods such as the Synthetic Minority Oversampling Technique (SMOTE), Edited Nearest Neighbours (ENN) and Random Oversampling (ROS), alone or combined into hybrid sampling (HS), are widely used to rebalance the classes [15, 16, 17]. How and when resampling is applied, however, determines whether the reported performance reflects the model's ability to recognise new patients. If the full dataset is resampled before it is split into training and test sets, duplicated and synthetic records derived from the same patient can appear on both sides of the split, and cleaning steps can remove difficult cases from the test set. The resulting estimates can be far higher than the performance a clinic would observe.

In order to improve CVD prediction models and methodologies, this research builds on recent studies. For example, recent studies investigated the effect of peripheral muscle action on cardio-exercise capacity in children with congenital heart defects [11], as well as fetal health monitoring with hybrid DL algorithms [12]. Furthermore, another innovative application of Blockchain and ML integration in ensuring healthcare data security holds immense potential for e-Healthcare solutions [13]. Our study combines DL techniques (MLP, CNN, NN) with ML algorithms (DT, SVM, XGBoost), feature selection and HS, and evaluates them under two protocols: the commonly used resample-before-split protocol and a leakage-free protocol in which the test data are held out before any sampling. A comprehensive list of abbreviations and their corresponding descriptions can be found in Table 1. The key contributions of this study are as follows:

- A pipeline for CVD prediction was implemented, incorporating data preprocessing, feature selection based on correlation, hybrid sampling (SMOTEENN followed by ROS) and

systematic evaluation of six DL and ML models.

- We show, quantitatively, that resampling the dataset before splitting it leaks information into the test set: all positive test records have identical copies in the training set and the ENN step removes 19.9% of the real negative records, which inflates accuracy to 94.7–99.8%.
- Under a leakage-free protocol we report the realistic performance of the same models on 6,000 real, unseen records (57.4–70.8% accuracy, AUC 0.57–0.81).
- We propose an improved, leakage-free pipeline (all features, class weighting, regularised models, validation-tuned decision threshold) that raises every model to about 76% accuracy, 80% recall and an AUC of 0.84, confirmed by five-fold cross-validation, and an ablation study that attributes the gain to each change.
- The complete code is released as open source so that every number and figure in this paper can be reproduced.

The primary goal of this research is to provide a reliable assessment of how well ML models can forecast CVD from survey data, and to give practical guidance on building and evaluating such models. The structure of this paper is as follows: Section 2 offers an overview of the related literature. Section 3 explains the methodology, including both evaluation protocols and the improved pipeline. Section 4 describes the models and evaluation metrics. Section 5 presents and discusses the results, and Section 6 concludes with a summary of the key findings and an analysis of their implications.

2 LITERATURE REVIEW

The literature reviewed in this study highlights key findings related to CVD prediction. One study proposed a solution for determining heart disease risk in highly imbalanced datasets using machine learning tools [14]. The authors investigated several approaches, including cost-sensitive learning algorithms, ensemble methods, and oversampling techniques. The experimental results demonstrated that ensemble methods and cost-sensitive learning algorithms were particularly effective in predicting CVD risk.

Building upon existing research, a method aimed at enhancing the survival prognosis of heart failure patients through the integration of SMOTE and advanced data mining techniques was proposed in [15]. This approach tackles the issue of class imbalance in heart failure datasets by generating synthetic instances for the underrepresented class using SMOTE. A range of classification algorithms, including KNN, NB, DT, XGBoost, and Random Forests were evaluated for heart failure prognosis, achieving prediction accuracy ranging from 85% to 94%. The study underscores the application of this methodology to improve both sensitivity and prediction performance for survival outcomes.

As highlighted in [16], data imbalance is a pervasive issue in classification tasks, and numerous strategies and algorithms have been proposed to mitigate its effects. The paper reviews the advantages and limitations of under-sampling, oversampling, and hybrid techniques. It presents a comprehensive overview of the existing approaches, stressing the importance of choosing the most suitable method based on the dataset's inherent characteristics. The authors conclude that addressing data imbalance in classification remains a critical area for further research and that the development of novel, more effective solutions is necessary.

Batista et al. [17] examined diverse methodologies for mitigating class imbalance in machine learning datasets. The analysis evaluated the efficacy of under-sampling, over-sampling, and hybrid strategies, including SMOTE combined with ENN, in enhancing model training. The outcomes underscored the critical role of balanced datasets in boosting classifier performance,

Table 1: List of Abbreviations

Abbreviation	Description
WHO	World Health Organization
CVD	Cardiovascular Disease
DL	Deep Learning
CKD	Chronic Kidney Disease
HRFLM	Hybrid Random Forest with a Linear Model
ML	Machine Learning
MLP	Multi-Layer Perceptron
CNN	Convolutional Neural Network
NN	Neural Network
DT	Decision Tree
SVM	Support Vector Machine
KNN	K-Nearest Neighbor
NB	Naive Bayes
LR	Logistic Regression
HDPM	Heart Disease Prediction Model
SMOTE	Synthetic Minority Oversampling Technique
ENN	Edited Nearest Neighbours
SMOTEENN	SMOTE followed by ENN cleaning
ROS	Random Oversampling
HS	Hybrid Sampling (SMOTEENN followed by ROS)
AUC-ROC	Area Under the Receiver Operating Characteristic Curve
CDC	Centers for Disease Control and Prevention
BRFSS	Behavioral Risk Factor Surveillance System
ReLU	Rectified Linear Unit
PPV	Positive Predictive Value
HS-MLP, HS-CNN, HS-NN	HS combined with MLP, CNN, NN
HS-DT, HS-SVM, HS-XGBoost	HS combined with DT, SVM, XGBoost

particularly in terms of precision and generalization. This study serves as a pivotal reference for employing hybrid sampling methods in contemporary machine learning applications, such as CVD prediction.

Extending the examination of feature selection and classification methodologies, Ali et al. [18] developed an effective machine learning framework for the early detection of cardiovascular diseases. Their methodology integrated advanced feature selection algorithms, including Relief and LASSO, with an ensemble of classification models to isolate the most clinically influential risk indicators and enhance diagnostic accuracy. The study demonstrated significant improvements in predictive sensitivity and specificity across medical records, underscoring the critical importance of feature optimization and balanced dataset evaluation for reliable clinical decision support.

A noteworthy study [19] investigates the implementation of ML techniques for detecting heart diseases within e-healthcare frameworks. The methodology combines conventional approaches with an ensemble of cutting-edge classifiers, such as logistic regression and KNN, to extract the most relevant features. When tested on a real-world dataset, the model delivered promising performance, achieving notable accuracy and precision metrics. This work highlights the potential of ML in advancing digital healthcare systems, particularly for the precise and timely diagnosis of cardiovascular conditions.

In the field of clinical decision support systems, Fitriyani et al. introduced the Heart Disease Prediction Model (HDPM) as a robust forecasting tool. Various machine learning algorithms, including SVM and KNN, were applied to construct a dependable and precise predictive model. Notably, HDPM surpassed other existing models, demonstrating an accuracy rate between 90% and 94% in diagnosing heart disease, with a sensitivity of 95% [20].

In furtherance of earlier research on ML for heart disease detection, a study was conducted

to investigate ML algorithms that could facilitate the swift diagnosis of CVDs [21]. This study assessed the performance of logistic regression, random forests, and SVM in detecting cardiac issues. Additionally, the influence of hyperparameter tuning through GridSearchCV on model performance was examined. The results indicated that improvements in algorithmic techniques and hyperparameter optimization contributed significantly to enhanced diagnostic accuracy in medical practice.

Ramesh et al. [22] focused on developing predictive models for heart diseases through the application of ML techniques. Logistic regression, decision trees (DT), and support vector machines (SVM) were employed for prediction. These models were assessed using real-world datasets, with performance evaluated based on precision, recall, and F1-scores. The results demonstrated that ML algorithms were effective in forecasting heart disease outcomes, with the authors emphasizing the significance of selecting an algorithm that aligns well with the specific characteristics of the data to ensure optimal performance.

A methodology to enhance the performance of mechanical circulatory support devices was introduced by Qureshi et al. [23]. This approach focuses on optimizing the transfer of kinetic energy from the forward-moving aortic blood flow. An exhaustive search technique was also employed to select the optimal cuff inflation and deflation timings. The feasibility and effectiveness of this method were explored through computer modeling and mathematical calculations. The findings suggest that optimizing kinetic energy transfer in mechanical circulatory support devices could significantly improve their functionality and increase cardiac output, presenting a new potential method to enhance the performance of the heart in patients with heart failure.

Azmi et al. [24] carried out a comprehensive review of AI applications in CVD prediction using large medical datasets. The objective of their review was to identify key trends and areas by conducting an extensive literature analysis. Among their discoveries, they illustrated how ML could be employed to effectively utilize big data for precise and efficient CVD prediction and early-stage diagnosis. The review highlighted the application of various ML algorithms, data pre-processing techniques, and evaluation metrics commonly used in CVD prediction studies. However, the authors emphasized that issues related to data quality, model interpretability, and scalability must be addressed before ML-based technologies can be widely implemented in the field of CVD prediction.

In a distinct study published in Diagnostics [25], a novel DNN model was introduced for the early identification and prediction of Chronic Kidney Disease (CKD). By incorporating Glomerular Filtration Rate (GFR) and kidney injury markers, this model surpassed traditional ML approaches, achieving perfect accuracy in classification tests. In a subsequent study, a hybrid parallel classification model for diagnosing CKD was developed, aiming to further enhance accuracy through innovative computational techniques [26]. A two-phase hybrid approach that integrates feature selection and Adaptive SVM for chronic disease classification was proposed in [27], and a further hybrid approach for chronic disease classification was presented in [28]. These studies explore the integration of multiple methodologies to improve the precision and efficiency of disease classification models.

One study utilized a mixed-method approach incorporating ML to achieve high prediction accuracy for heart diseases [29]. By integrating a hybrid random forest with a linear model (HRFLM), the predictive accuracy of the heart disease model was significantly enhanced. The researchers reported an accuracy rate of 88.4%, outperforming individual algorithms. In a separate effort to forecast CVD, Kavitha et al. [30] introduced a hybrid prediction method integrating Random Forest and DT, which achieved an accuracy of 88.7%.

Several authors have warned that preprocessing steps applied before data partitioning, including oversampling, can leak information from the test set into model training and inflate reported performance [40, 41]. Santos et al. [42] showed in particular that oversampling before cross-validation produces over-optimistic estimates on imbalanced data, and recommended that resampling be applied only to the training folds. These findings motivate the two evaluation

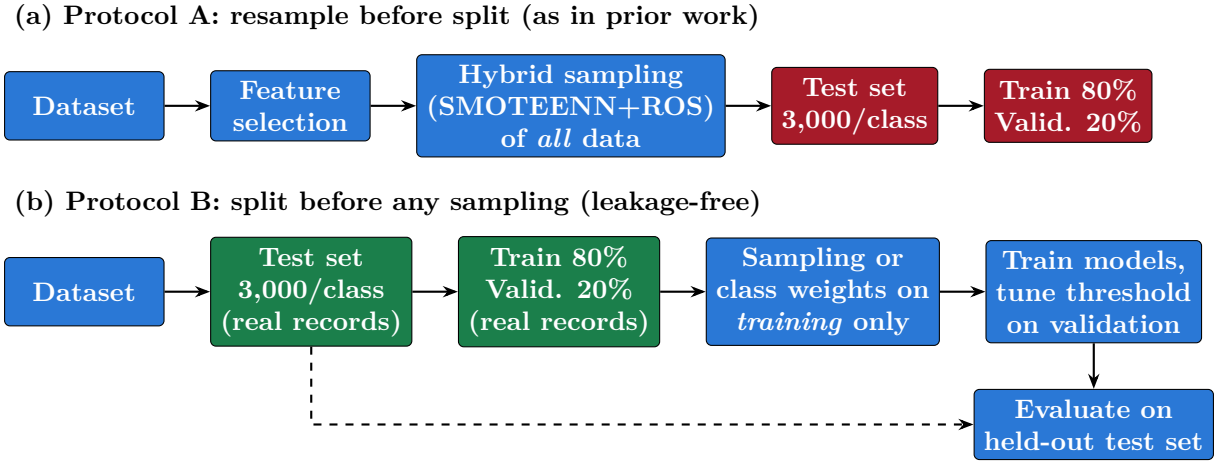


Figure 1: Proposed methodology. (a) In the resample-before-split protocol used by the original pipeline, synthetic, duplicated and cleaned records are created before the test set is drawn. (b) In the leakage-free protocol the test and validation sets contain only real records that are never resampled or seen during training.

protocols compared in this study.

The methodologies reviewed in this study represent substantial advancements in CVD prediction. Collectively, these studies indicate a promising trajectory for improving CVD prediction accuracy and early detection. However, the reported accuracies are obtained on different datasets and, often, under different evaluation protocols, which makes them difficult to compare. A balanced evaluation that separates genuine predictive power from evaluation artefacts is necessary to guide future efforts.

3 PROPOSED METHODOLOGY

This section outlines the methodology for CVD prediction. The data analysis is implemented in Python using pandas, scikit-learn, imbalanced-learn, XGBoost and TensorFlow/Keras. The overall pipeline is shown in Fig. 1; its key design choice is the point at which sampling is applied relative to the train/test split, which defines the two evaluation protocols compared throughout the paper.

3.1 Data Collection

The dataset employed in this study was sourced from Kaggle [31], a publicly accessible data repository. It is a refined version of the data originally compiled by the Behavioral Risk Factor Surveillance System (BRFSS) [32]. The dataset consists of 253,680 records, each representing responses from the 2015 BRFSS survey conducted by the Centers for Disease Control and Prevention (CDC). It includes 22 columns, one of which is the target variable, "HeartDiseaseorAttack". This binary target variable indicates the presence or absence of a heart attack, where "0" signifies no history of coronary heart disease (CHD) or myocardial infarction (MI) and "1" indicates a reported history of CHD or MI. The raw data comprise 229,787 records without heart disease and 23,893 records with heart disease. A comprehensive description of the dataset, including the variables and their corresponding measurements, is provided in Table 2 [33].

3.2 Data Preprocessing

This study employs an exploratory data analysis (EDA) approach. Upon loading the dataset using the Pandas library, duplicate entries are removed to maintain data integrity, leaving 229,781

Table 2: Description of Dataset Variables

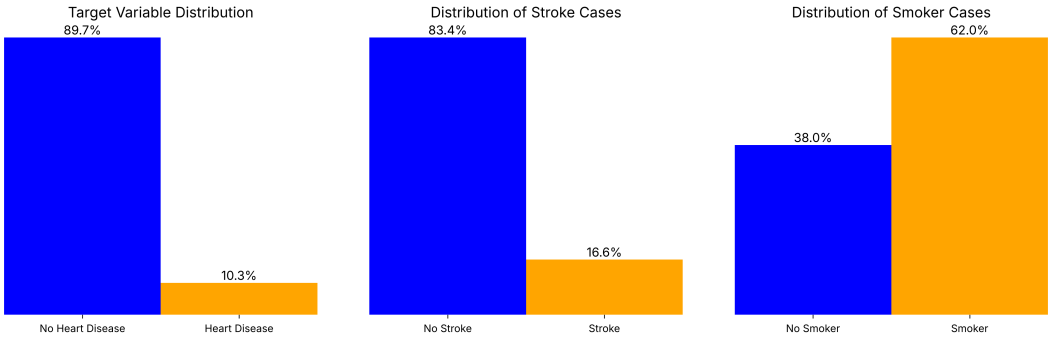
Variable	Description
HeartDiseaseorAttack	0 = No history of coronary heart disease (CHD) or myocardial infarction (MI); 1 = Reported CHD or MI
HighBP	0 = No high blood pressure; 1 = High blood pressure
HighChol	0 = No high cholesterol; 1 = High cholesterol
CholCheck	0 = No cholesterol check in past 5 years; 1 = Checked cholesterol in past 5 years
BMI	Body Mass Index [12–98]
Smoker	0 = Smoked fewer than 100 cigarettes in lifetime; 1 = Smoked at least 100 cigarettes in lifetime
Stroke	0 = Has never experienced a stroke; 1 = Has experienced a stroke
Diabetes	0 = Absence of diabetes; 1 = Prediabetes / borderline; 2 = Presence of diabetes
PhysActivity	0 = No physical activity or exercise in prior month; 1 = Engaged in physical activity or exercise in prior month
Fruits	0 = Consumes fruit less than once per day; 1 = Consumes fruit one or more times per day
Veggies	0 = Consumes vegetables less than once per day; 1 = Consumes vegetables one or more times per day
HvyAlcoholConsump	Adult men: $\geq$14 drinks/week; Adult women: $\geq$7 drinks/week (0 = No, 1 = Yes)
AnyHealthcare	Health care coverage / health insurance (0 = No, 1 = Yes)
NoDocbcCost	Could not see doctor due to cost in past 12 months (0 = No, 1 = Yes)
GenHlth	General health rating: [1, 2, 3] = Excellent, Very Good, Good; [4, 5] = Fair, Poor
MentHlth	Days of poor mental health in past 30 days [0–30]
PhysHlth	Days of physical illness or injury in past 30 days [0–30]
DiffWalk	Serious difficulty walking or climbing stairs (0 = No, 1 = Yes)
Sex	0 = Female, 1 = Male
Age	13-level category: 1 = 18–24, 2 = 25–29, 3 = 30–34, 4 = 35–39, 5 = 40–44, 6 = 45–49, 7 = 50–54, 8 = 55–59, 9 = 60–64, 10 = 65–69, 11 = 70–74, 12 = 75–79, 13 = 80 or older
Education	[1, 2, 3] = Did not graduate high school; 4 = High school graduate; 5 = Attended college or technical school; 6 = College graduate
Income	1, 2 = Lower than \$15,000; 3, 4 = \$15,000 to less than \$25,000; 5 = \$25,000 to less than \$35,000; 6 = \$35,000 to less than \$50,000; 7, 8 = \$50,000 or more

unique records with 22 columns. The presence of missing data is assessed, and the analysis confirms that no missing values exist. The target variable is strongly imbalanced: the dataset comprises 23,717 cases of heart disease (10.3%) and 206,064 instances of individuals without heart disease.

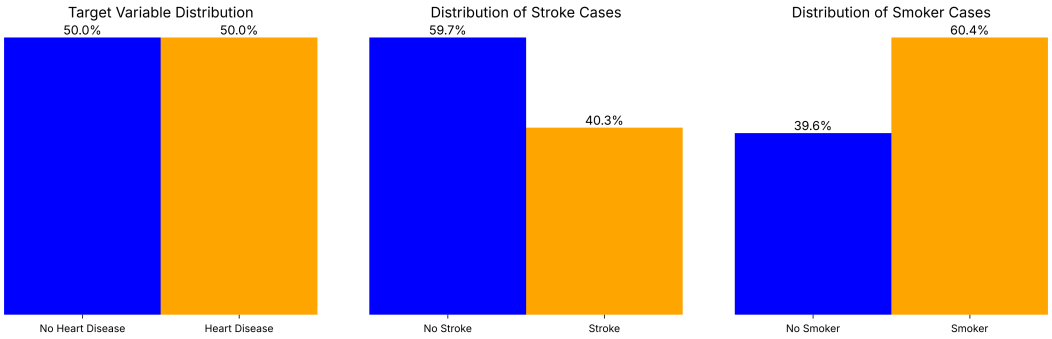
Fig. 2(a) depicts the distribution of the target variable and, among the heart disease cases, the proportions of individuals who have had a stroke or who smoke. The age distribution of individuals with and without heart disease is compared in Fig. 3(a); the prevalence of heart disease increases steeply with age.

3.3 Feature Selection

Feature selection helps reduce the dataset's complexity while focusing on the variables most closely linked to the target outcome. A Pearson correlation matrix is computed and the features are ranked by their correlation with "HeartDiseaseorAttack". The strongest associations are



(a) Before sampling



(b) After hybrid sampling of the whole dataset (Protocol A)

Figure 2: Distribution of the target variable, and of stroke and smoker cases among individuals with heart disease, before and after hybrid sampling.

found for GenHlth ($r = 0.246$), Age (0.224), DiffWalk (0.203), HighBP (0.201), Stroke (0.199), HighChol (0.176), Diabetes (0.171), PhysHlth (0.171) and Smoker (0.105). Following the original pipeline, these nine features are used by the HS models. Because Pearson correlation only captures linear, univariate association, the improved pipeline (Section 3.6) instead uses all 21 features and lets the models weigh them; the effect of this choice is quantified in the ablation study.

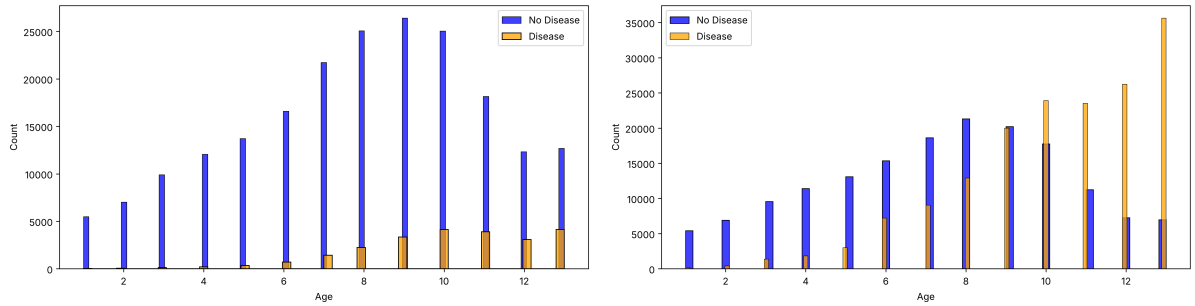
3.4 Hybrid Sampling

Hybrid sampling combines SMOTEENN with ROS. SMOTE first synthesises new minority-class instances by interpolating between a minority record and its nearest minority neighbours until the classes are balanced. ENN then removes every record whose class disagrees with the majority of its three nearest neighbours, from both classes. Finally, ROS duplicates minority records until both classes have equal size. The HS methodology is implemented with the imbalanced-learn library (random state 42).

3.5 Evaluation Protocols

Protocol A (resample before split). This is the protocol of the original pipeline. HS is applied to the whole dataset, which yields 330,288 records (165,144 per class). From this resampled dataset 3,000 records per class are drawn at random as the test set; the remaining records are split 80/20 into training (259,430) and validation (64,858) sets. Five-fold cross-validation is likewise run on the resampled data.

Protocol B (split before sampling, leakage-free). Before any sampling, 3,000 *real* records per class are drawn from the deduplicated dataset as the test set. The remaining records are



(a) Before sampling **(b)** After hybrid sampling (Protocol A)

Figure 3: Age distribution of individuals with and without heart disease before and after hybrid sampling.

split, stratified by class, into training (179,024 records, 80%) and validation (44,757 records, 20%) sets. Sampling, when used, is applied to the training set only; the validation and test sets contain only real records with their natural characteristics. The test set is balanced, so its accuracy equals the balanced accuracy, which allows a direct comparison with Protocol A. For cross-validation the 223,781 non-test records are split into five stratified folds; the held-out fold keeps the real prevalence of about 10.3%, and any sampling, early stopping and threshold tuning use only the training folds.

3.6 Improved Pipeline

Based on the analysis in Section 5, we propose an improved pipeline that is evaluated under Protocol B. It makes four changes:

1. **All 21 features** are used instead of the nine correlation-selected ones.

2. **Class weighting replaces resampling.** The models are trained on the real, imbalanced training data; errors on the minority class are weighted by the inverse class frequency (`class_weight="balanced"` for DT, SVM, LR and the DL models; `scale_pos_weight` = 9.8, the ratio of negative to positive training records, for XGBoost).

3. **Regularised models.** The DT is limited to a depth of 8 with at least 50 records per leaf; XGBoost uses shallow trees (depth 5), learning rate 0.03, row and column subsampling of 0.8 and early stopping on the validation AUC (patience 100 rounds, at most 2,000 trees); the DL models use early stopping on the validation loss (patience 3, at most 30 epochs, batch size 256) and restore the best weights. Logistic regression (LR) is added as an interpretable baseline.

4. **Decision threshold tuned on the validation set.** Instead of the default threshold of 0.5, the threshold that maximises Youden's index $J = \text{sensitivity} + \text{specificity} - 1$ on the validation set is used.

4 DL AND ML ARCHITECTURE ANALYSIS

4.1 ML Models

Decision Tree. A DT classifier (C4.5-style, entropy splitting criterion) learns a hierarchy of feature tests that map input features to the CVD label. In the HS pipeline the tree is grown to full depth.

SVM. An SVM classifier with a linear kernel separates the two classes with a maximum-margin hyperplane. Features are min–max scaled, and the model is fitted with the LIBLINEAR solver, which solves the same linear SVM problem as a kernel SVM with a linear kernel but scales to several hundred thousand records. Its decision values are used to compute the ROC curve.

XGBoost. XGBoost builds an ensemble of gradient-boosted trees, each correcting the errors of the previous ones. The HS pipeline uses the library defaults.

4.2 DL Models

All DL models share a preprocessing stage: features are min–max scaled and passed through an autoencoder whose encoder (one dense layer of 64 ReLU units, trained for 10 epochs to reconstruct its input with a sigmoid output layer and mean-squared-error loss) produces a 64-dimensional representation that is fed to the classifier. The classifiers are:

- **CNN**: Conv1D (8 filters, kernel size 2, ReLU) → max pooling (2) → flatten → dense (16, ReLU) → dense (1, sigmoid); 20 epochs.
- **NN**: dense (64, ReLU) → dense (32, ReLU) → dense (1, sigmoid); 10 epochs.
- **MLP**: dense (128) → dense (64) → dense (32), all ReLU → dense (1, sigmoid); 20 epochs.

All DL models are trained with the Adam optimiser and binary cross-entropy loss, with a batch size of 32 in the HS pipeline.

4.3 Evaluation Metrics

For performance evaluation, a confusion matrix is employed to analyze the results of each model. The matrix produces four key metrics: true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN). These metrics facilitate the computation of additional performance indicators.

1. Accuracy: the proportion of correct predictions [34]:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{1}$$

2. Precision (positive predictive value): the proportion of predicted CVD cases that are correct [35]:

$$\text{Precision} = \frac{TP}{TP + FP} \tag{2}$$

3. Recall (sensitivity): the proportion of actual CVD cases that are detected [35]:

$$\text{Recall} = \frac{TP}{TP + FN} \tag{3}$$

4. F1-score: the harmonic mean of precision and recall [34]:

$$\text{F1-score} = \frac{2 \cdot (\text{Precision} \cdot \text{Recall})}{\text{Precision} + \text{Recall}} \tag{4}$$

Table 3: Test performance under Protocol A (resample before split; 3,000 test records per class drawn from the resampled data). Last column: accuracy reported in the original version of this study.

Sampling	Model	Acc. (%)	Prec. (%)	Rec. (%)	F1 (%)	AUC (%)	Orig. Acc. (%)
SMOTEENN	CNN	88.58	96.81	79.80	87.48	98.08	74.09
	NN	89.18	96.41	81.40	88.27	98.38	75.08
	MLP	91.15	97.72	84.27	90.50	99.12	75.00
ROS	CNN	75.55	72.70	81.83	77.00	83.17	75.13
	NN	75.47	72.63	81.73	76.91	83.20	75.51
	MLP	75.63	72.39	82.87	77.28	83.30	75.91
HS	CNN	96.02	93.13	99.37	96.15	98.73	93.01
	NN	94.70	94.49	94.93	94.71	98.63	95.96
	MLP	98.45	97.12	99.87	98.47	99.47	97.55
	DT	99.78	99.57	100.00	99.78	99.78	99.70
	SVM	89.57	88.79	90.57	89.67	95.96	93.16
	XGBoost	98.23	96.80	99.77	98.26	99.65	97.80

5. Area Under the Curve (AUC): the area under the receiver operating characteristic (ROC) curve, which traces the true positive rate (TPR) against the false positive rate (FPR) across classification thresholds [34]. The AUC ranges from 0.5 (random guessing) to 1.0 (perfect classification) and is computed from the continuous model scores:

$$\mathrm{AUC} = \int_0^1 \mathrm{TPR}(\mathrm{FPR})\, d(\mathrm{FPR}) \tag{5}$$

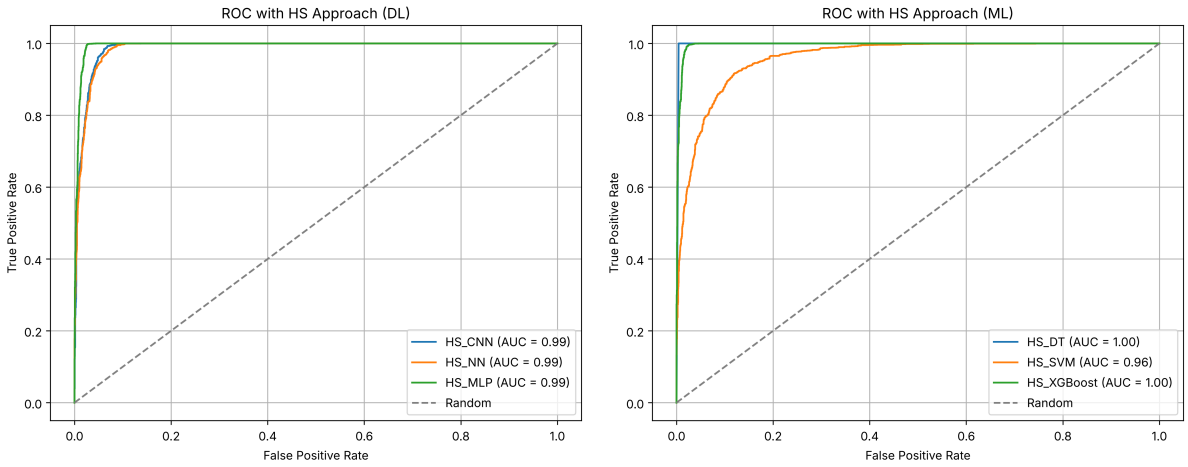
Unlike accuracy, precision and F1, the AUC does not depend on the decision threshold or on the class balance of the test set, which makes it the most robust metric for comparing models across protocols.

5 RESULTS AND DISCUSSION

The results are presented in four steps. Section 5.1 reports the performance of the HS models under the resample-before-split protocol (Protocol A). Section 5.2 analyses why these results do not transfer to new patients. Section 5.3 evaluates the same models and the improved pipeline on real, held-out records (Protocol B), including an ablation study and five-fold cross-validation, and Section 5.4 compares the findings with existing studies. All experiments use a fixed random state of 42 and can be reproduced with the released code.

5.1 Performance under the Resample-Before-Split Protocol

Table 3 reports the test-set performance of all models when the full dataset is resampled before it is split (Protocol A), as in the original pipeline. With hybrid sampling every model appears highly accurate: the DT reaches 99.78% accuracy and 100% recall, XGBoost 98.23%, the MLP 98.45% and the CNN and NN 96.02% and 94.70%; the ROC curves (Fig. 4) hug the top-left corner with AUC values of 0.96–1.00. Models trained with ROS alone, in contrast, reach only about 75.5% accuracy and an AUC of 0.83. The last column lists the accuracies reported in the original version of this study; our reimplementation reproduces them closely for HS and ROS. For SMOTEENN alone our reimplementation obtains higher values than originally reported (88.6–91.2% vs. 74–75%), consistent with the effect of ENN cleaning on the test set discussed in Section 5.2.



(a) HS with DL models

(b) HS with ML models

Figure 4: ROC curves under Protocol A (resample before split). The near-perfect curves are an artefact of data leakage (Section 5.2).

Table 4: Overlap between test and training sets for hybrid sampling: percentage of test records that have an identical record (all nine features and the label) in the training set. Negative records overlap even without leakage because healthy respondents often give identical answers to the nine mostly binary questions.

Protocol	All test records	Positive (CVD)	Negative
A: resample before split	98.5%	100.0%	96.9%
B: split before sampling	40.9%	4.6%	77.1%

5.2 Data Leakage in the Resample-Before-Split Protocol

Two properties of HS explain why the Protocol A results overstate how the models would perform on new patients.

Duplication. SMOTE first creates 182,347 synthetic positive records to balance the classes. ENN then removes most of the positive records together with 40,920 real negative records, leaving 22,239 positive and 165,144 negative records. ROS finally duplicates each remaining positive record about 7.4 times on average to balance the classes. When the test set is drawn from these data, its positive records are copies of records that also occur in the training set. Table 4 quantifies this: under Protocol A, 100% of the positive test records have an identical record (same features and label) in the training set, compared with 4.6% under Protocol B. A fully grown decision tree can simply memorise these records, which explains its near-perfect score.

Cleaning of the test population. ENN removes every record whose label disagrees with most of its nearest neighbours. These are exactly the patients whose survey answers look like those of the opposite class: 19.9% of all real negative records are deleted before the test set is drawn. The test set therefore no longer contains the borderline cases that make CVD prediction from survey data hard, and the classes become artificially separable. This is why HS raises even the smooth DL models from about 75.5% (ROS alone, which removes no records) to 94.7–98.5%. In a clinical setting such patients cannot be removed.

Both effects are well-documented forms of data leakage [40, 41]: information about the test data enters model training or shapes the test data themselves. Santos et al. [42] showed that oversampling before cross-validation produces over-optimistic estimates, and our results quantify

Table 5: Test performance under Protocol B (6,000 real records, 3,000 per class, held out before any sampling). Left: original HS pipeline (nine features, hybrid sampling of the training data, threshold 0.5). Right: improved pipeline (all 21 features, class weights, regularised models, threshold tuned on the validation set). All values in %.

	Original HS pipeline					Improved pipeline				
Model	Acc.	Prec.	Rec.	F1	AUC	Acc.	Prec.	Rec.	F1	AUC
DT	57.40	83.23	18.53	30.32	57.40	74.90	73.06	78.90	75.87	82.50
SVM	70.78	76.39	60.17	67.31	81.06	75.98	73.89	80.37	76.99	83.67
XGBoost	61.58	80.67	30.47	44.23	74.31	76.00	72.90	82.77	77.52	83.99
CNN	66.85	76.45	48.70	59.50	77.58	75.55	73.43	80.07	76.61	83.41
NN	64.48	78.31	40.07	53.01	74.67	75.88	74.46	78.80	76.57	83.85
MLP	63.05	80.90	34.17	48.04	72.53	75.47	72.37	82.40	77.06	83.79
LR	–	–	–	–	–	76.00	73.93	80.33	77.00	83.70

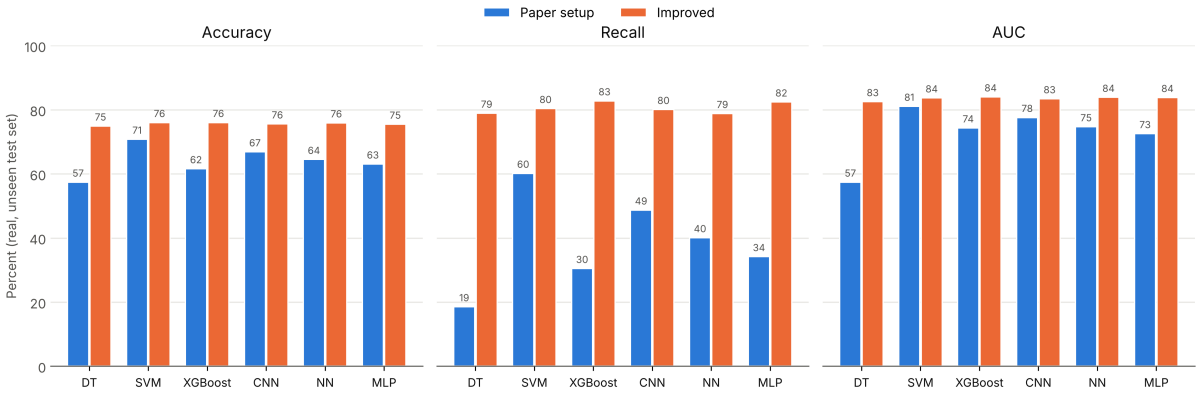


Figure 5: Accuracy, recall and AUC on the real, unseen test set (Protocol B) for the original HS pipeline and the improved pipeline.

the size of this bias for CVD prediction: more than 40 percentage points of accuracy for the DT. The cross-validation results of the original version of this study (92.7–99.7% accuracy for the HS models) suffer from the same problem, because the folds were drawn from the resampled data, and are therefore replaced by leakage-free cross-validation in Section 5.3.4.

5.3 Performance on Real, Unseen Records

5.3.1 The original HS pipeline under Protocol B

Table 5 (left) shows the same HS models trained on the resampled training set and evaluated on 6,000 real records that were held out before sampling. Performance drops sharply: accuracy falls to 57.4–70.8% and the AUC to 0.57–0.81. The fully grown DT, the best model under Protocol A, is close to chance level (AUC 0.574) because the records it memorised do not recur among new patients. All HS models are also very conservative on real data: their recall ranges from 18.5% (DT) to 60.2% (SVM), so most CVD cases are missed. Because ENN removed the borderline negatives from the training data, the models learn that only clear-cut profiles are negative and place the default decision threshold of 0.5 far from where it would be on the real population.

5.3.2 Ablation study

To identify which design choices limit performance on real data, we varied them one at a time for XGBoost and the DT (Table 6). The AUC measures how well a model ranks patients by

Table 6: Ablation study under Protocol B (real test set). "Tuned" denotes the threshold that maximises Youden's J on the validation set.

Model	Configuration	AUC (%)	Acc. @ 0.5 (%)	Acc. @ tuned (%)
XGBoost	9 features, HS (original)	74.31	61.58	68.28
XGBoost	9 features, ROS only	82.06	74.35	74.25
XGBoost	9 features, no resampling	82.25	53.27	74.55
XGBoost	21 features, no resampling	83.54	54.75	75.25
XGBoost	21 features, class weights, regularised (improved)	83.99	–	76.00
DT	9 features, HS, full depth (original)	57.40	57.40	57.40
DT	21 features, class weights, depth 8 (improved)	82.50	–	74.90

risk, independently of the threshold; accuracy is reported both at the default threshold of 0.5 and at the threshold tuned on the validation set.

Four findings emerge. First, hybrid sampling is the main cause of the loss: replacing it with ROS alone raises the AUC of XGBoost from 0.743 to 0.821, and dropping resampling altogether gives the same AUC (0.823), showing that synthetic records and ENN cleaning distort the data rather than add information. Second, without resampling the default threshold of 0.5 is unsuitable for imbalanced data (53.3% accuracy, because almost every record is predicted negative), whereas a threshold tuned on the validation set restores 74.6% accuracy; resampling and threshold tuning address the same problem, but tuning does so without altering the data. Third, using all 21 features instead of the nine correlation-selected ones increases the AUC from 0.823 to 0.835: features such as sex, income, BMI and education carry predictive information that a univariate correlation ranking misses. Fourth, regularisation (shallow, early-stopped boosting and a depth-limited tree) prevents the models from memorising individual records; it turns the DT from a near-random classifier into a competitive model (AUC 0.825).

5.3.3 The improved pipeline

Table 5 (right) and Fig. 5 show that the improved pipeline raises every model on real, unseen records. Accuracy increases to 74.9–76.0%, recall to 78.8–82.8% and the AUC to 0.825–0.840. The largest gains concern recall: the improved XGBoost detects 82.8% of CVD cases, compared with 30.5% for the original HS XGBoost, and the improved DT detects 78.9% compared with 18.5%. The ROC curves of all improved models nearly coincide (Fig. 6), and logistic regression performs on par with XGBoost and the three DL models. The confusion matrices (Fig. 7) show a balanced error profile: the improved XGBoost correctly identifies 2,483 of 3,000 CVD cases and 2,077 of 3,000 non-CVD cases. The learning curves of the DL models (Fig. 8) show the validation loss decreasing for about ten epochs and then levelling off; training stops three epochs after the last improvement and the best weights are restored, which prevents over-fitting. The training loss is computed with class weights and is therefore not directly comparable to the unweighted validation loss.

5.3.4 Five-fold cross-validation

To check that these results do not depend on one particular split, the improved models were evaluated with stratified five-fold cross-validation on the 223,781 real records that are not in the test set. In every fold, 20% of the training portion was set aside for early stopping and threshold tuning, and the held-out fold kept the real prevalence of about 10.3%. Table 7 reports the mean and standard deviation over the folds. The AUC (0.824–0.839) and recall (78.9–82.5%) agree with the test-set results and vary little between folds (standard deviation of the AUC at most 0.64 percentage points), confirming that the estimates are stable. Accuracy (70.8–72.2%) is

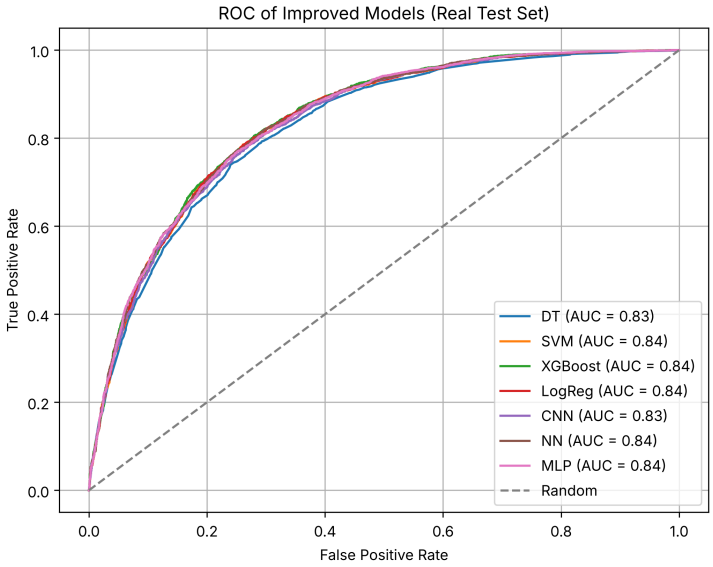
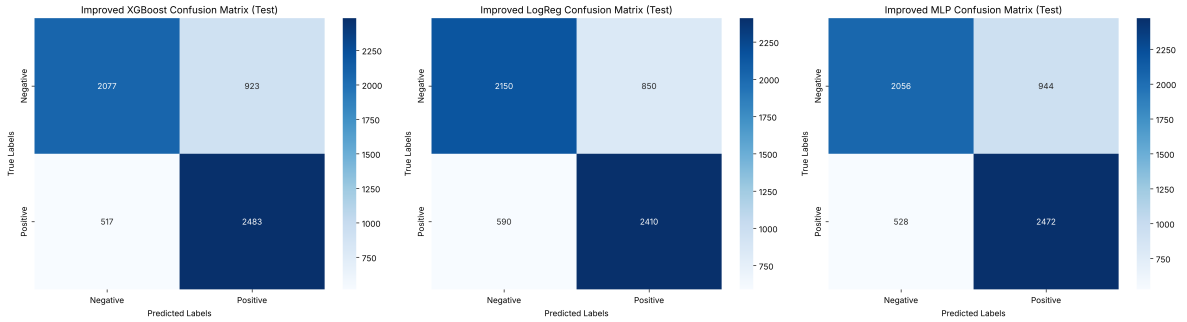


Figure 6: ROC curves of the improved models on the real, unseen test set (Protocol B).



(a) XGBoost **(b)** Logistic regression **(c)** MLP

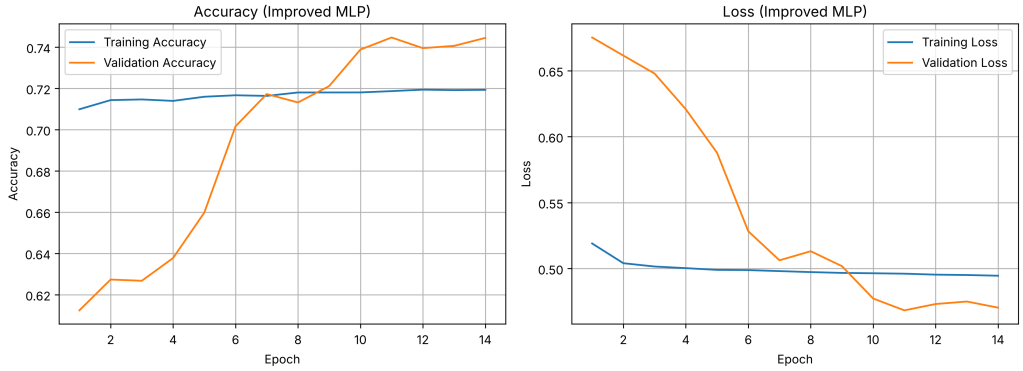
Figure 7: Confusion matrices of improved models on the real test set (Protocol B, 3,000 records per class).

lower than on the balanced test set because the tuned threshold trades specificity for sensitivity and negatives make up almost 90% of each fold.

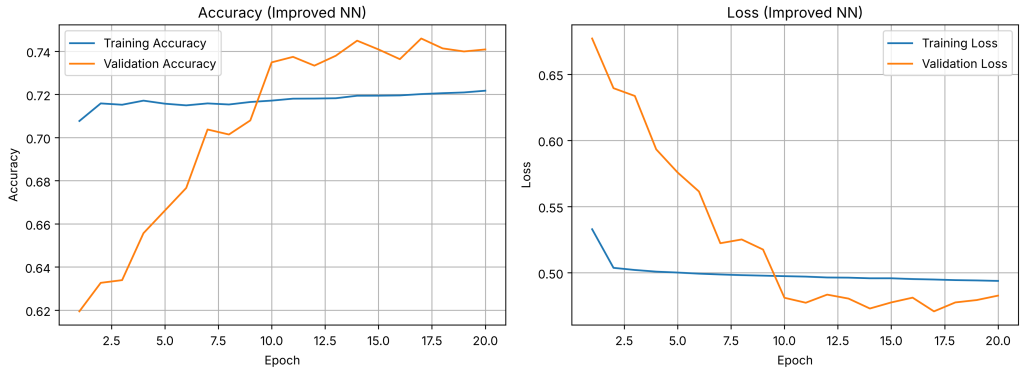
Precision and F1 are much lower in cross-validation than on the balanced test set because the folds keep the real prevalence. At a prevalence of 10%, even a model with about 80% sensitivity and 70% specificity flags roughly three to four healthy respondents for every true CVD case (precision about 22%). This is an inherent property of screening for a condition of low prevalence, not a defect of a particular model, and it should guide how such models are used: as inexpensive pre-screening tools that refer at-risk individuals for clinical assessment, not as diagnostic tests.

5.4 Comparison with Existing Studies

Table 8 compares the proposed method with existing studies. With a few exceptions, previous studies achieved accuracy rates between 82% and 96%, and studies on small datasets such as the Cleveland heart disease data (303 records) report the highest values. These figures are not directly comparable: the studies use different datasets with different clinical variables, and several do not state whether resampling or feature selection was performed before the data were split. Our results show that this choice alone can change the reported accuracy of the same model on the same data from 57% to 99.8%. For the proposed method we therefore report



(a) MLP



(b) NN

Figure 8: Training and validation accuracy and loss of the improved DL models; training stops early once the validation loss no longer improves.

the leakage-free results on 6,000 real BRFSS records. Of the studies listed, Amarbayasgalan et al. [39] used the KNHANES survey (25,340 records), which is closest in nature to BRFSS.

5.5 Limitations

The BRFSS data are self-reported and cross-sectional: the target records a past diagnosis of coronary heart disease or myocardial infarction, so the models identify individuals who resemble patients with a history of CVD rather than predict future events. Self-reported survey answers also carry less information than clinical measurements, which limits the attainable discrimination. The models were developed and tested on a single survey year without external validation, and the calibration of the predicted probabilities was not assessed. The balanced test set makes accuracy comparable across protocols, but precision and F1 depend on prevalence, and the cross-validation results indicate the values to expect in a general population.

6 CONCLUSION AND FUTURE WORK

This study combined hybrid sampling (SMOTEENN followed by ROS) with three DL models (CNN, NN, MLP) and three classical ML models (DT, SVM, XGBoost) for CVD prediction from the 2015 BRFSS survey, and evaluated them under two protocols. When the full dataset is resampled before it is split, the models report 94.7–99.8% test accuracy. We showed that this performance is an artefact of data leakage: every positive test record has an exact copy in the training set, and the ENN step removes a fifth of the real non-CVD records, the borderline cases, before the test set is drawn. On 6,000 real records held out before any sampling, the same

Table 7: Five-fold cross-validation of the improved models on real records at the natural prevalence of CVD (about 10.3%); mean ± standard deviation over the folds, in %.

Model	Accuracy	Precision	Recall	F1 Score	AUC
DT	71.82 ± 1.92	21.84 ± 0.98	78.90 ± 2.17	34.18 ± 1.01	82.36 ± 0.36
SVM	72.16 ± 0.57	22.28 ± 0.36	80.65 ± 0.49	34.92 ± 0.45	83.53 ± 0.38
XGBoost	71.08 ± 1.00	21.88 ± 0.49	82.50 ± 1.22	34.58 ± 0.55	83.85 ± 0.35
LR	71.38 ± 1.68	21.94 ± 0.82	81.52 ± 1.72	34.56 ± 0.90	83.55 ± 0.38
CNN	71.41 ± 1.41	21.88 ± 0.73	81.04 ± 1.54	34.44 ± 0.84	83.25 ± 0.64
NN	70.79 ± 1.09	21.61 ± 0.52	81.95 ± 2.05	34.20 ± 0.61	83.37 ± 0.51
MLP	72.19 ± 2.54	22.29 ± 1.28	80.01 ± 2.49	34.83 ± 1.37	83.36 ± 0.38

Table 8: Comparison of the proposed method with existing studies. The proposed results are leakage-free (Protocol B, real held-out test set).

Study	Dataset	Method	Acc. (%)	Prec. (%)	Rec. (%)	F1 (%)	AUC (%)
M. Bhatt et al. [36]	CVD dataset	XGBoost	87.02	89.62	82.11	86.30	95.00
S. K. Mohan et al. [29]	Cleveland	Hybrid (HRFLM)	88.40	90.10	92.80	90.00	–
N. L. Fitriyani et al. [20]	Cleveland	SMOTEENN + XGBoost	95.90	97.14	94.67	95.35	–
W. Chang et al. [37]	Beijing dataset	Hybrid XGBSVM	91.70	–	–	–	93.90
A. A. Almazroi et al. [38]	Cleveland	DNN	82.49	–	87.24	84.41	–
J. P. Li et al. [19]	Cleveland	SVM (Linear)	85.00	–	75.00	–	–
T. Amarbayasgalan et al. [39]	KNHANES	Two DNNs	89.20	91.10	92.00	91.50	97.54
Proposed (improved)	BRFSS	XGBoost	76.00	72.90	82.77	77.52	83.99
Proposed (improved)	BRFSS	Logistic regression	76.00	73.93	80.33	77.00	83.70
Proposed (improved)	BRFSS	MLP	75.47	72.37	82.40	77.06	83.79

models reach only 57.4–70.8% accuracy, and the fully grown decision tree that appeared best falls to near chance level.

We therefore proposed a leakage-free pipeline that uses all available features, handles class imbalance with class weights instead of synthetic records, regularises the models and tunes the decision threshold on a validation set. On real, unseen records it improves every model, to about 76% accuracy, 80% recall and an AUC of 0.84, values that five-fold cross-validation confirms. The technical observations of this study are:

- The evaluation protocol dominates the reported performance. Any resampling, cleaning or duplication must be applied after the test data have been set aside, and only to the training data; otherwise accuracy on imbalanced clinical data can be overstated by more than 40 percentage points.
- On real data, hybrid sampling did not improve discrimination: it lowered the AUC of XGBoost from 0.823 to 0.743. Class weighting and a tuned decision threshold address imbalance without distorting the data.
- Correlation-based selection of nine features discarded useful information; using all 21 features raised the AUC.
- Model choice matters far less than data handling. Logistic regression, SVM, XGBoost and the three DL models all reach an AUC of 0.83–0.84, so simple and interpretable models are a sound choice for this task.
- At the real prevalence of about 10%, a model that detects 80% of CVD cases still produces three to four false alarms for every true case (precision about 22%). Survey-based models are therefore suited to low-cost screening and risk stratification that triggers clinical follow-up, not to diagnosis.

Future work should validate the pipeline on external cohorts and later BRFSS survey years, study the calibration of the predicted probabilities, and add clinical measurements (for example blood pressure readings, lipid profiles and ECG data), which are likely needed to move beyond the performance ceiling of self-reported data. Explainability methods such as SHAP values could make the models' risk factors transparent to clinicians, and threshold selection could be aligned with the costs of missed cases and false alarms in a specific screening programme. We also encourage authors of CVD prediction studies to report the point at which resampling is applied and to release their code, so that results can be verified and compared.

CODE AND DATA AVAILABILITY

The dataset used in this study is publicly available on Kaggle [31] and in the GitHub repository at `https://github.com/daniyalabdulah/Early-Detection-of-Heart-Disease-using-Deep-Learning`. The complete source code, which reproduces every table and figure in this paper, is available at `https://github.com/mohsinkhan85/Biomedical-Neurocomputing-Lab`.

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